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Abstract

In a world where communication defines connection, not everyone has an equal voice. For individuals who are mute, expressing simple needs like "I need help" or "I'm in pain" can become daily

challenges, especially when those around them do not understand sign language. This communication gap often leads to frustration, social isolation, and dependency on others.

Our project, MUTE, bridges this gap through an integrated smart system built around a wearable

glove equipped with flex sensors and powered by the ESP32 microcontroller. The glove detects the bending of each finger, capturing the unique signature of hand gestures and translating them into meaningful text instantly and accurately.

To ensure reliable interpretation, we collected real-world data from 20 participants, covering over 23 sign gestures, and trained a customized machine learning model on this dataset. The model achieved an impressive 98% accuracy, demonstrating its ability to generalize across different

users and gestures.

Complementing the glove is a full-featured Flutter mobile application designed with accessibility and simplicity in mind. The app enables:

- .Real-time Bluetooth communication with the glove via Serial Bluetooth.
- .Instant gesture translation using an integrated TensorFlow Lite model.
- .Text-to-Speech (TTS) output in both male and female Arabic voices.
- .Bilingual support (Arabic-English) for inclusive communication.
- .User authentication via email sign-up, login, and Google account integration.
- .A dedicated “Transcribe” screen that displays recognized words live.
- .Extra utilities: Profile settings, notifications, app settings, and support/help section.

Unlike generic solutions, MUTE is deeply human-centered, it empowers users to express themselves naturally without requiring the other party to know sign language. It's lightweight, wireless,

responsive, and supports real-time interaction.

In the next phases, MUTE could evolve to support gesture-based control, context-aware predictions, or facial emotion recognition, unlocking even more possibilities in secure, contactless, and adaptive human-computer interaction systems.

MUTE is more than a graduation project, it’s a meaningful step toward a world where every voice is seen, heard, and understood.

1. Introduction

1.1 Overview

In an era defined by technological innovation and a commitment to inclusivity, smart gloves have emerged as a revolutionary solution, reshaping communication possibilities for individuals with speech impairments. These innovative wearables represent a convergence of cutting-edge hardware and intelligent software, designed to empower users by providing an intuitive and independent means of expression.

Smart gloves go beyond mere technological advancement; they mark a transformative leap in addressing the communication challenges faced by people especially those individuals who struggle to express themselves verbally.

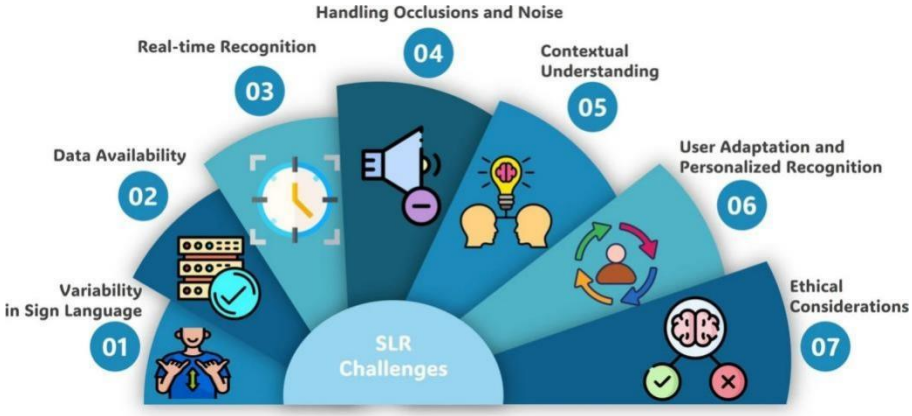
These devices, typically comprising a strategic combination of flex sensors and Bluetooth connectivity, stand as a testament to the power of innovation in fostering autonomy and inclusion. As the gloves seamlessly translate intricate sign language gestures into text or speech, they not only break down communication barriers but also challenge the societal stigmas often associated with speech disabilities.

1.2 Problem Definition

Despite advancements in technology, millions of individuals with speech impairments continue to face fundamental obstacles in expressing their needs and emotions in public and private life. Communication remains a critical barrier, especially in communities where sign language is not commonly taught or understood, which can lead to social marginalization, isolation, and limited access to personal relationships, career opportunities, and education.

1.2.1 Several factors contribute to this communication gap:

1. **Language barriers:** Sign language is the primary mode of communication for many mute individuals. However, it is not widely understood by the hearing population. This gap leaves mute people excluded from everyday conversations and important decision-making processes.
2. **Inaccessible environments:** Many public spaces, workplaces, and educational institutions lack support for sign language, making it difficult for mute people to participate fully and equally.
3. **Limited opportunities:** Communication barriers often lead to missed chances for employment and access to higher education, limiting personal and professional growth.
4. **Social isolation:** The inability to communicate effectively can result in feelings of loneliness, disconnection, and difficulty forming meaningful relationships.



1.2.2 The absence of live translation creates major barriers:

1. **Mutual Unintelligibility:** Most of the hearing population lacks knowledge of sign language, leading to a disconnect between mute individuals and the surrounding community.
2. **Lack of Inclusive Tools:** Current assistive solutions are either too bulky, rely on visual input (like cameras), or require both parties to adapt, which can be impractical in dynamic real-life situations.
3. **Emergency Inaccessibility:** In urgent scenarios such as medical emergencies, the inability to express basic needs like “I need help!” or “I

feel pain!” can have serious consequences.

4. **Loss of Autonomy:** Relying on interpreters or written communication strips mute individuals of spontaneous and independent interaction, especially in fast-paced or unfamiliar environments.

These challenges reveal the need for a portable, intuitive, and discreet assistive communication system, one that empowers users without requiring others to adapt. This is precisely where MUTE finds its purpose.

1.3 Solution

The MUTE system bridges the communication gap by integrating smart wearable hardware with an intuitive mobile application, offering a comprehensive assistive solution for the deaf and hard- of-hearing community.

The system enables real-time interaction through the following key features:

✧ **Sign Language Translation:**

The smart glove captures finger movements using flex sensors and sends them via Bluetooth to the mobile app, where a machine learning model interprets the gestures and translates them into text and voice.

✧ **Text-to-Speech Output:**

Recognized gestures are vocalized using natural-sounding male or female voices, allowing deaf or mute users to express themselves verbally in real-time.

✧ **Bilingual Translation:**

The app supports both Arabic and English, making communication accessible in diverse environments.

✧ **User-Friendly App Integration:**

Built with Flutter, the app includes features such as sign-up/login (with Google integration), real-time translation, user profiles, notifications, and settings.

This integrated approach empowers users to communicate independently, confidently, and inclusively, reshaping how non-verbal individuals interact with the world around them.

1.4 Project Motivation

At the heart of this project lies a simple yet powerful belief: **technology should serve humanity**. We are driven by the desire to create solutions that make a real impact—helping individuals overcome barriers, especially those that affect something as fundamental as communication.

Being a human is about meeting new people, making new memories, and being with the ones who care about you, and you care about them. Imagine that you can help someone and give him a method to overcome his disability.

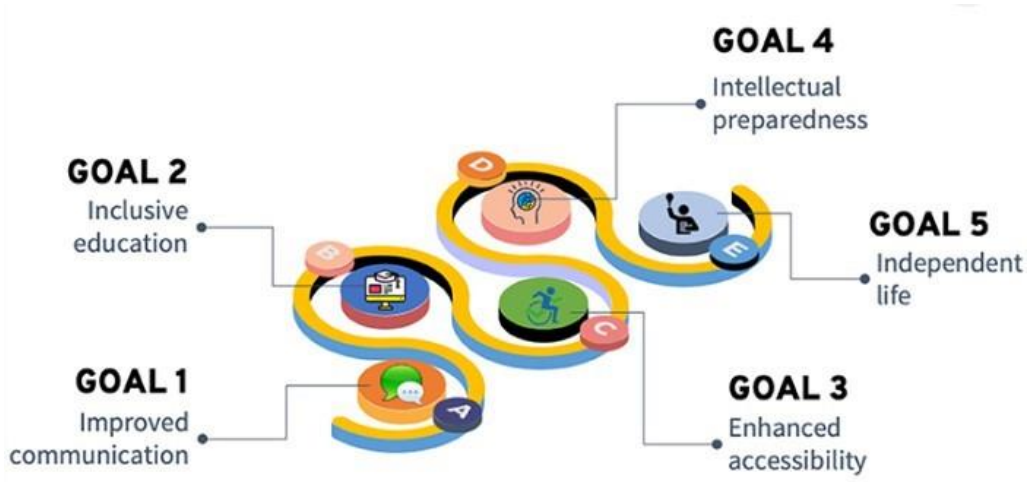
We aim to give people the tools they need to express themselves freely, to stay connected with their families, loved ones, and society.

This project is not just about innovation, it's about restoring dignity, independence, and human connection. Whether it helps a father communicate with his children or allows a patient to express pain or needs to a caregiver, the potential impact goes beyond technology. Through this initiative, we hope to make the world a little more inclusive, compassionate, and empowering for everyone.

1.5 Project Objectives

With a mission to make communication more accessible, this project combines hardware and software to empower individuals with speech disabilities to express themselves effortlessly.

1) Develop an end-to-end communication system that enables mute individuals to express themselves clearly and independently in real-time.
2) Empower users to communicate without the need for interpreters or writing by achieving the following:
    a) Design a low-cost smart glove equipped with flex sensors to detect hand gestures and convert them into meaningful text and speech outputs.
    b) Develop a user-friendly mobile application that connects to the glove via Bluetooth, displays the recognized signs in real time, and provides additional functionalities such as:
        i) Voice output (Text-to-Speech)
        ii) Language translation
        iii) User authentication
        iv) Real-time gesture transcription
3) Ensure wireless, seamless interaction between the glove and the app to make communication as natural and effortless as possible.
4) Deliver the solution within a defined timeline, using available academic and technical resources effectively.



1.6 Organization

This documentation is structured into several chapters to cover all stages and components of the project:

Chapter 2, Literature Review

Presents an overview of existing systems, prior academic work, and related technologies in the field of sign language recognition and assistive communication devices.

Chapter 3, Project Planning

Describes the initial planning phase, including task breakdown, timeline, resource allocation, and the rationale for choosing the proposed system.

Chapter 4, Development Methodology and System Specification

Explains the adopted Agile development methodology, system architecture, and detailed specifications including both functional and non-functional requirements.

Chapter 5, Embedded System

Focuses on the hardware component of the project, including the selection of ESP32, sensor integration, PCB design, firmware development, and Bluetooth communication protocols.

Chapter 6, Machine Learning Model

Covers dataset collection, preprocessing, model architecture, training, validation, accuracy metrics, and deployment of the trained model for gesture recognition.

Chapter 7, Flutter Mobile Application

Details the design and implementation of the mobile application using Flutter, including BLE integration, UI/UX design, TFLite model integration, and accessibility features.

Each chapter is designed to provide clear insight into its respective component while demonstrating how all modules work together to deliver a complete and functional solution.

1.7 Marketing

To ensure widespread adoption and accessibility, the marketing plan aims to scale

the project from a local initiative to national, regional, and eventually international markets. The strategy is composed of the following key elements:

1.7.1 Digital Advertising and Online Presence

Launch targeted **sponsored ads** on platforms such as **YouTube**, focusing on users with speech disabilities.

Develop a **dedicated website** for the product, offering:

- Detailed product information and specifications
- Step-by-step usage guides and video tutorials
- Online ordering with **nationwide and international shipping** options
- Customer support and maintenance services

1.7.2 Social Media Engagement

Leverage platforms like **Facebook, Instagram, and TikTok** to raise awareness and engage directly with potential users.

Target **niche communities** and **accessibility-focused groups** to reach individuals who would benefit most from the solution.

1.7.3 Institutional Collaboration

Partner with **charitable organizations, schools for the mute,** and **rehabilitation centers** to promote the device and reach end users directly.

Offer **bulk purchase options** for institutions working with the speech impaired communities.

1.7.4 Public Demonstrations and Outreach

Conduct **live demonstrations**, **workshops**, and **awareness events** to showcase the product's features, usability, and impact.

Participate in **technology expos, health conferences,** and **community events** to build credibility and visibility.